



CLEAN FUTURE
INTERNATIONAL



VOLUME III • SEPTEMBER 2026

CFI WASH Sanitation Hub

Audit and redesign of the business model

The existing pitch board rebuilt on the CFI 2.0 architecture — lifecycle, approval gates, the three money layers and validated unit economics.



THE QUESTION THIS DOCUMENT ANSWERS

Not "can CFI build a toilet block?" but "at what level of use does one keep itself working for twenty years — and who pays if it doesn't?"

NAVIGATE THIS VOLUME

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SOURCE AND STATUS

Built from the CFI Miro board *Clean Future International Business Model*. Every figure originates there; the arithmetic has been reworked and is presented as a **model**, not a projection. Nothing here is verified until the footfall study in Section 11 is complete.

Board frames that were empty — Team, Quote, Competitive analysis and Feedback — are not carried over.

THE FINDING IN ONE LINE

The board's model works — but only above roughly 71 paying users a day. Below that it is social infrastructure needing subsidy, and the board does not say so.

01

What the Board Gets Right

The audit starts here because the board's core insight is correct, and the redesign preserves it rather than replacing it.

Read the board as a first draft of an investment case rather than as a finished one and it holds up well. The diagnosis is sound, the physical specification is competent, and the commercial instinct is right. What it lacks is the discipline that turns a good idea into something a trustee can approve.



It diagnoses maintenance failure, not scarcity

"Existing facilities break down due to: no maintenance plan, no cleaning budget, no ownership structure, no accountability." That is the exact thesis of CFI 2.0, arrived at independently from the field rather than from a framework.



It names an operator from the start

The on-site caretaker system and a caretaker salary line appear in the model itself. Most WASH proposals leave the operator unnamed and unfunded; this one does not.



It builds a revenue mechanism into the asset

User fees, kiosk rental and advertising panels are designed in, not bolted on — which is what makes a Type A or B classification possible at all.



It specifies the physical system, not just the building

Gender-separated blocks, handwashing, borehole and water storage, solar lighting, septic or biodigester. These are the components that decide whether a facility still works in year three.



It has real before-and-after evidence

The proven-concept frame carries site photography rather than renders. That is the beginning of a baseline, and it is worth far more to a funder than the visualisations.



It states a hybrid financial identity

"Hybrid NGO + Social Enterprise" is the right shape, and matches the hybrid model in Volume I Section 14.

None of this needs rebuilding. What follows is not a disagreement with the model — it is the difference between a pitch that describes an idea and an investment case that survives due diligence.

02

What Has to Change

Seven structural issues. The first is the one that produces most of the others.

None of these require abandoning work already done. Five are additions, one is a re-sequencing, and one is tidying.

1

The chain starts at Build

"CFI CONNECTS: Build → Operate → Maintain → Scale." Four stages, beginning with construction. Volume I's lifecycle has twelve and begins at Discover and Verify. Starting at Build is precisely the conventional approach the board's own problem statement criticises.

2

It is a pitch template, not an architecture

Problem · Solution · How it works · Business model · Team · Quote · Timeline · Competitive analysis · Feedback. That is a startup deck. Four of those frames are still empty, and the structure cannot express gates, divisions, layers or lifecycle.

3

Revenue is presented before it is validated

Footfall of 100 and 500 adults per day is assumed, not measured. Volume II is explicit: a revenue assumption untested in comparable operating conditions is an estimate and must be labelled as one.

4

There is no renewal provision

Operating costs cover caretaker, cleaning and utilities. Nothing sets aside the cost of replacing the asset. A model that cannot fund its own replacement is not sustainable — it is deferring the problem by one asset life.

5

The organization has no funding line

Five roles are named — Regional Director, Finance Officer, Operations Manager, Community Liaison Officer, Site Supervisors — and no income source reaches them. This is the Layer 3 gap from Volume II, visible in the model itself.

6

No gates, no measurement, no safeguarding

Nothing states what must be true before construction is approved, how impact will be measured, or how children and vulnerable users are protected at a facility sited in schools.

7

Presentation defects

Template placeholders still present ("One Line Slide Title / Your Name"); an unexplained figure 845650 floating on the canvas; photographs in .HEIC, which will not display for most funders; and two competing taglines — "Building Today. Sustaining Tomorrow." on the board against "Engineering better futures. Sustaining them." in Volume I.

03

The Numbers That Do Not Hold

Four arithmetic problems, all of them fixable, and one of them serious enough to change the project's classification.

The board contradicts itself on the same revenue figure

WHERE IT APPEARS	MONTHLY	IMPLIED BASIS
Revenue simulation frame	GHS 6,200	100 users × GHS 2 × 31 days
Business model frame	GHS 6,000	100 users × GHS 2 × 30 days
High-traffic estimate	GHS 31,000	500 users × GHS 2 × 31 days

A funder who spots this before you do will question every other number on the board. Both conventions are also wrong in the same direction: a sanitation facility does not earn at full rate every calendar day. The corrected model in Section 09 uses 365 days at a stated utilisation, so the assumption is visible rather than buried in a month length.

Stated operating costs do not reconcile

LINE	MONTHLY	ANNUAL
Caretaker salary	1,200	14,400
Cleaning supplies	600	7,200
Water and utilities	500	6,000
Total as listed	2,300	27,600
Board's stated operating cost	—	30,000–40,000

The gap of GHS 2,400–12,400 is unexplained. It is probably right — desludging, repairs, consumables, security and bank charges are real — but an unexplained gap reads as guesswork. Section 09 itemises it.

The serious one. Capital is GHS 250,000–400,000 and annual surplus at low traffic is roughly GHS 21,000 once renewal is provided for. That is a **12 to 19 year payback** — longer than the asset's likely working life. At low traffic this is not a revenue-sustainable project, and the board presents it as one.

04
The Verified Problem

The board's problem statement, restated so that each claim carries the evidence CFI would need at Gate 1.

Nothing in this table disputes the board's claims — they are almost certainly true. The point is that "almost certainly true" is not what Gate 1 asks for, and the evidence needed is cheap to gather.

CLAIM ON THE BOARD	EVIDENCE HELD	EVIDENCE STILL REQUIRED AT GATE 1
Basic, JHS and SHS schools lack proper toilet facilities	Site photography, six locations	Enrolment numbers, pupil-to-facility ratio, absence data by gender, district education office confirmation.
Public spaces lack hygienic toilets	Site photography, two locations	Footfall counts at candidate sites — the same data that validates revenue in Section 11.
Facilities break down: no maintenance plan, no cleaning budget, no ownership, no accountability	Field observation	Structured condition survey of existing facilities in the district; who owns each, who funds cleaning, when it last failed.
Open defecation and poor hygiene affect school attendance	Asserted	The weakest link in the chain. Needs district health or education data, or a CFI baseline survey — it cannot rest on assertion.

8

Sites photographed across schools and public spaces

4

Named causes of failure — all institutional, not technical

0

Claims currently supported by measured baseline data

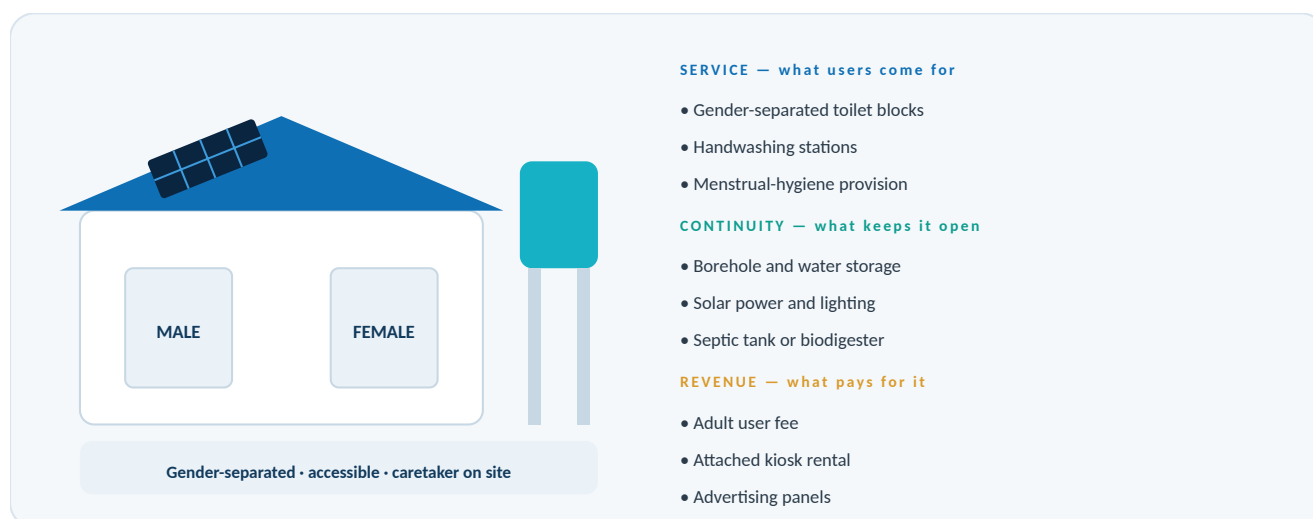
Why this matters more than it looks. Every one of the four causes the board identifies is institutional — a missing plan, budget, owner or accountability line. None is engineering. That is the strongest argument for the CFI model, and it should lead the case rather than sit in a corner of the problem frame.

05

The Sanitation Hub

The board's build specification, grouped by what each component is *for* — service, continuity, or revenue. The final design is always site-specific.

Grouping matters because the three categories fail differently. Service components fail visibly and get fixed; continuity components fail silently — a dry borehole or a dead solar battery closes a hub weeks before anyone reports it.



Where hubs are sited

Basic, JHS & SHS

Highest need, lowest ability to pay

Markets & community centres

High footfall, revenue viable

Bus and lorry stations

Highest footfall of all

Leisure areas

Seasonal, needs testing

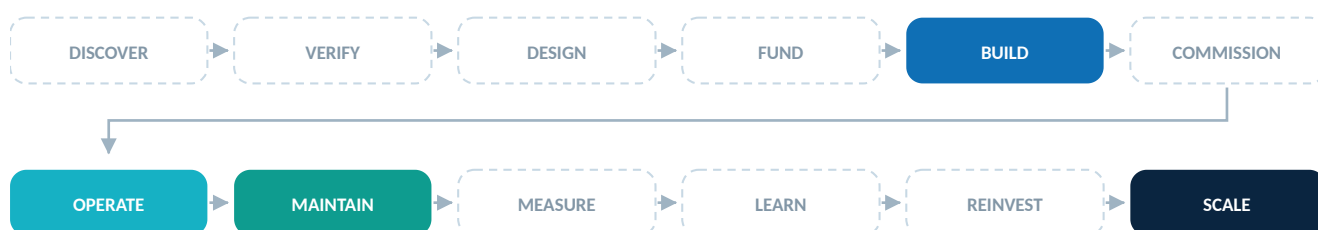
These four sites do not share one financial model. A school hub has almost no paying users; a lorry station may have five hundred a day. Treating them as one product with one revenue line is the central modelling error to fix — see Section 09.

06 Where It Sits in the Lifecycle

The board's chain — Build, Operate, Maintain, Scale — is four of CFI's twelve stages.

The eight it omits are where projects of this kind normally fail. Three of them come *before* construction, which is why a chain that starts at Build cannot protect itself: by the time the first block is laid, the decisions that determine whether it survives have already been made or skipped.

SHADED = ON THE BOARD • OUTLINED = MISSING



The four that decide whether this project is fundable

Verify Turns observation into evidence a funder can check.	Fund Requires the operating plan funded, not just the build.	Commission Defects closed, operator trained, handover signed.	Measure Proves the hub still works — and funds the next one.
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The board's own diagnosis argues for the missing stages. It says facilities fail for want of a maintenance plan, a cleaning budget, an owner and accountability. Those four things are decided at Fund and Commission — two of the stages the chain skips.

07

Gate Status of This Project

Assessed against the twelve approval gates in Volume I. This is what a trustee or funder would ask for, in the order they would ask.

A gate is not passed because work has started on it. It is passed when the evidence exists and someone has signed to say so.

GATE	STATUS	WHAT IS OUTSTANDING
0 • Problem	Passed	A meaningful problem is clearly established and locally observed.
1 • Evidence	Partial	Photography is not verification. Needs enrolment data, condition survey and footfall counts.
2 • Feasibility	Partial	Technically credible. Land tenure, district permissions and water-source yield unconfirmed.
3 • Sustainability	Not passed	Operator model exists on paper; no maintenance schedule, no renewal reserve, no signed operator agreement.
4 • Finance	Not passed	Capital not raised; operating plan not funded; revenue unvalidated. This is the binding gate.
5 • Design	Not passed	Architectural prototype listed as a next step. No independent technical review yet.
6–11	Not reached	Procurement, construction, commissioning, operations, impact and scale follow in sequence.



Progress stops at Gate 3. Everything in Part C exists to get Gates 3 and 4 open.

THE HONEST POSITION

This is a Gate 2 project being presented as a Gate 5 project. That is normal at this stage and not a criticism — but saying so openly is what separates a credible case from an optimistic one.

08

The Three Money Layers, Applied

The board has one pot of money. Volume II requires three, because they have different sources and different failure modes. Applied to a single hub:

The separation is not bookkeeping pedantry. Each layer has a different funder, a different time horizon and a different consequence when it runs dry — and money raised for one usually cannot lawfully be spent on another.

Layer 1

PROJECT CAPITAL

Building the hub — GHS 250,000 to 400,000

Land preparation, foundation and structure, plumbing, septic tank or biodigester, solar installation, water tank, tiling and finishing, handwashing stations, caretaker office, branding.

Source: grants, CSR, institutional capital. Ends at commissioning.

Layer 2

HUB OPERATING

Keeping it working — roughly GHS 48,000 a year

Caretaker, cleaning supplies, water and utilities, desludging, repairs, and the renewal provision that replaces the asset at end of life.

Source: user fees, kiosk rental, advertising — and subsidy where those fall short.

Layer 3

CFI CORE

Running the organization — currently unfunded

Regional Director, Finance Officer, Operations Manager, Community Liaison Officer, Site Supervisors. Plus governance, audit, safeguarding and the registries.

Source: unrestricted income and cost recovery. Neither appears anywhere on the board.

Layer 2 is the one the board comes closest to getting right, and the one most WASH proposals omit entirely — but its renewal line is missing, which is what Section 09 adds.

The caretaker is funded; the Operations Manager who supervises the caretaker is not. That is the whole Layer 3 problem in one sentence. Five named roles carry no income line, so in practice they would be paid out of project money — which most grant agreements prohibit, and which Volume II Section 11 names as a red line.

09 Unit Economics, Corrected

One hub, one year, all figures in Ghana cedis. Built from the board's own assumptions with the gaps filled in and stated. **model, not a projection**

Annual cost of keeping one hub working

COST LINE	MONTHLY	ANNUAL	SOURCE
Caretaker salary	1,200	14,400	Board
Cleaning supplies	600	7,200	Board
Water and utilities	500	6,000	Board
Direct operating cost	2,300	27,600	
Renewal provision	1,667	20,000	Added — GHS 300,000 over 15 years
Organizational cost recovery	345	4,140	Added — 15% per Volume II
Full annual cost	4,312	51,740	

Desludging, corrective repairs and security are not yet itemised and belong in direct operating cost. Adding them is what closes the board's unexplained gap between GHS 27,600 and its stated GHS 30,000–40,000.

Annual revenue at the board's two traffic assumptions

SCENARIO	USERS/DAY	REVENUE	FULL COST	SURPLUS
Low traffic — school or small community	100	73,000	51,740	21,260
High traffic — market or lorry station	500	365,000	~90,000	~275,000

At GHS 2 per adult use, 365 days. The high-traffic column assumes costs rise with use — a second caretaker, more consumables, more frequent desludging — which the board does not currently allow for.

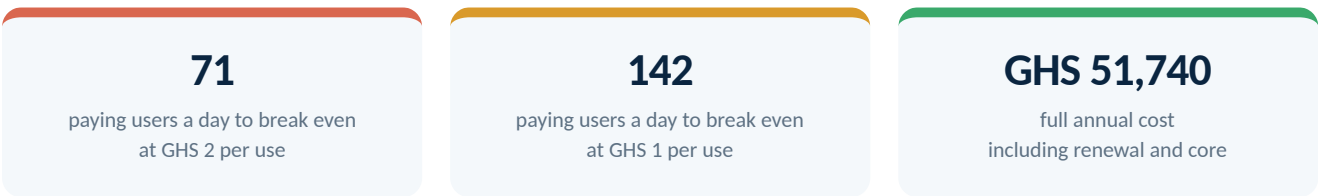
Capital payback tells the real story. At low traffic the surplus of GHS 21,260 repays GHS 250,000–400,000 of capital in **12 to 19 years** — longer than the asset lasts. At high traffic it repays in **under two years**. Same building, entirely different investment.

SECTION 09 CONTINUED

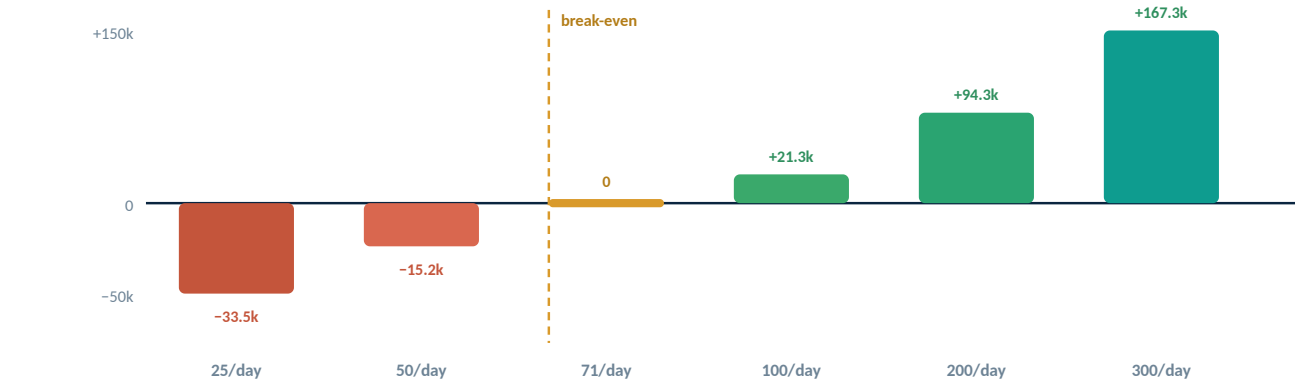
Break-Even and Sensitivity

One number decides whether a site is viable, and it is not the construction cost. It is daily paying users.

Break-even here means covering everything: operating cost, the renewal provision that replaces the asset, and the hub's share of running CFI. It is a higher bar than the board's operating-cost-only comparison, and it is the bar a funder will apply.



Surplus or deficit by daily footfall at GHS 2 per use



Annual surplus after operating cost, renewal provision and organizational cost recovery.

What this changes about site selection. A school hub will rarely reach 71 paying users a day — pupils cannot be charged. School sites are therefore subsidised by design, and should be funded as such. Market and lorry-station hubs can carry themselves and, above roughly 150 users a day, can cross-subsidise a school hub. That is a far stronger proposition than "toilets generate revenue", and it is what the footfall study in Section 11 must establish.

10

Sustainability Category

Volume I requires every project to be classified before funding. The board implies Type A. The evidence supports Type B, and Type C for schools.

Crucially, the category belongs to the *site*, not to the programme. One CFI WASH portfolio will contain all three, and each needs its own funding structure.

A

Revenue-Sustainable
Lorry stations and large markets only, and only once footfall above ~150/day is measured rather than assumed.

B

Partially Revenue-Supported
Community centres and smaller markets. Real income, but grants or subsidy still required — particularly for renewal.

C

Social Infrastructure
Every school hub. Pupils are not charged, so there is no revenue line at all. The operating model must be explicitly subsidised.

What the classification changes

IF THE HUB IS	THEN THE FUNDING ASK MUST INCLUDE
Type A	Capital only. The hub funds its own operation, renewal and a share of CFI's core cost.
Type B	Capital, plus a stated multi-year subsidy for the shortfall — named, quantified and time-bound.
Type C	Capital, plus the <i>full</i> operating and renewal cost for the asset's life, or a committed institutional owner (district, school authority) who carries it.

Why this is the most important page here. Funding a Type C asset with a Type A business case is how facilities end up abandoned — money for the building, nothing for the twenty years afterwards. It is the exact failure the board's own problem statement describes, and classification is what prevents CFI repeating it.

11 Validating the Revenue

Volume II forbids putting an unvalidated revenue assumption into a final investment case. This is the study that removes that objection — and it is cheap.

1

Count, don't estimate

Two weeks of manual footfall counts at three candidate sites, at three times of day, weekday and weekend. Record total passers-by and those who actively seek a facility.

2

Test willingness to pay, not just presence

A short intercept survey: would you pay GHS 1? GHS 2? What do you currently do instead? Presence is not demand — the existing free alternative is the real competitor.

3

Benchmark against operating facilities

Paid public toilets already operate in Ghanaian markets and stations. Their actual daily takings are the single best predictor available, and the operators can usually be asked directly.

4

Price the secondary revenue honestly

Kiosk rental and advertising are asserted on the board with no figures. Obtain one real quote for each before either appears in a funding document.

5

Classify the site, then design to it

The count determines Type A, B or C — which determines the funding ask, the operating model and whether the site proceeds at all.

Cost

Two enumerators, three sites, two weeks. Small relative to a GHS 250,000 build.

Timing

Runs inside months 3–6 of the 24-month plan, before Gate 4.

Effect

Converts the strongest objection to this model into its strongest evidence.

WORTH SAYING PLAINLY

If the count comes back at forty users a day, that is not a failed study — it is the study working. It means the site is Type C and must be funded accordingly, which is far better discovered now than in year three.

Organization and the Core-Cost Gap

The board names five roles. Mapped onto the CFI divisions, they cover the operating half of the model well and the front half barely at all.

This is a sensible team for running hubs that already exist. It is not yet a team that can find, verify, design and fund new ones.

ROLE ON THE BOARD	CFI DIVISION	LAYER	COMMENT
Regional Director	Leadership	3	Unfunded
Finance Officer	Shared — Finance	3	Unfunded. Required for Gate 4 and for any grant.
Operations Manager	CFI Operations	3	Unfunded, yet supervises the funded caretaker.
Community Liaison Officer	CFI Community	3	Unfunded. This is the Discover and Verify capability.
Site Supervisors	CFI Delivery	1 or 3	Chargeable to a project while construction runs; unfunded between projects.
Caretaker (<i>per hub</i>)	CFI Operations	2	Funded — GHS 1,200/month in the model.

Divisions with no named role at all

CFI Engineering

No technical lead. The architectural prototype and design review have no owner.

CFI Capital

Nobody owns fundraising, yet Layer 3 depends entirely on unrestricted income.

Impact & Digital

No owner for baselines, functionality monitoring or the registries.

The fix is the 15% recovery rate. Each hub's budget should carry roughly GHS 4,140 a year toward these roles, and each construction grant a share of the build-phase overhead. One hub funds a fraction of one salary — which is precisely why Volume II insists unrestricted income is raised first, before the first build is committed.

13

What Happens Next

The board's five next steps, re-sequenced so that each one unlocks the next and none commits money before the evidence exists.

The order is the substance of the change. The board's sequence raises money first and gathers evidence afterwards; this one reverses that, because evidence is what makes the money raisable at all.

0-3

MONTHS

Fix the model and the institution

Correct the revenue arithmetic. Adopt lifecycle costing with a renewal line. Set the cost-recovery rate. Register the Ghana branch. Assemble the due-diligence pack.

Unlocks: a funding case that survives scrutiny.

3-6

MONTHS

Measure, then classify

Run the footfall and willingness-to-pay study at three candidate sites. Complete the condition survey and baseline. Classify each site Type A, B or C. Commission the architectural prototype and have it independently reviewed.

Unlocks: Gates 1, 2 and 5.

6-12

MONTHS

Fund one hub properly, not three cheaply

Select the single strongest site. Raise Layer 1 with recovery included and Layer 2 with its renewal provision, as two separate asks. Sign the operator agreement before construction.

Unlocks: Gates 3 and 4 — and construction.

12-24

MONTHS

Prove it with data, then replicate

Commission, operate, and publish real functionality, footfall, revenue and cost figures. Replace every modelled number in this document with a measured one.

Unlocks: the case for hubs two and three — built on evidence.

One change from the board's plan. It proposes seed funding for three facilities. Fund one. Three unproven hubs multiply an unvalidated assumption by three; one hub, measured for a year, turns it into evidence — and evidence is what pays for the next ten.



CLEAN FUTURE INTERNATIONAL

CFI WASH — Sanitation Hub

Every figure in this document is modelled from the CFI Miro board's own assumptions to show how the economics behave. None is verified. The footfall study in Section 11 is what converts this from a model into an investment case.

Three things to fix on the board itself. Remove the template placeholders and the unexplained figure 845650; convert the .HEIC photographs to JPEG so funders can open them; and settle on one tagline — "Engineering better futures. Sustaining them." is the one carried through Volumes I and II.

Document control. CFI WASH — Sanitation Hub Investment Case, Volume III. Issued September 2026. Sections 01–13, Parts A–D. Source: CFI Miro board *Clean Future International Business Model*. Companion to Volume I (Strategic Framework & Master Model) and Volume II (Funding & Sustainability Model). Not financial, engineering or tax advice — costings, revenue assumptions and the renewal basis require professional review and confirmation with CFI's auditor before use in a funding decision.